PRODUCT USER MANUAL

For Arctic Ocean BGC Reanalysis Product

ARCTIC_MULTIYEAR_BGC_002_005

Issue: 1.0

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RECORD TABLE

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1.0	2025/05/28	All	Initial version transferred from PUM all	C. Yumruktepe, T. Wakamatsu, L. Bertino, S. Gao, H. Sumata	L. Bertino

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations

it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

ARCTIC_ANALYSISFORECAST_BGC_002_005 is the nominal product of the Arctic Monitoring and Forecasting Center for ocean biogeochemistry, and is composed of 3D, daily mean fields of nitrate, phosphate, silicate, oxygen, chlorophyll, phytoplankton biomass, primary production, and light attenuation.

b) Polar projections

All products are interpolated onto the same polar stereographic output grid, although at different resolutions. HYCOM uses a staggered Arakawa C grid but all variables are defined at the center of the output grid cell after interpolation.
The same projection is used for all Arctic MFC grids on a spherical Earth, following this definition under the proj4 projection library: (Beware, this is not the WGS84 ellipsoid reference).

projection_stereo:proj4 = "+units=m +proj=stere +a=6378273.0 +b=6378273.0 +lon_0=-45.0 +lat_0=90.0 +lat_ts=90.0 +ellps=sphere" ;

This polar stereographic projection also defines the reference direction of vectors (current and sea ice velocity), which should therefore be rotated if the data is reprojected to a Lat-Long or any other grid.
In ArcGIS and QGIS softwares, the x- and y- coordinates units should be converted from 100 km to meters.

c) Vertical coordinates

All Arctic ocean products – except the sea ice product - are computed on a hybrid vertical coordinate. This is a combination of isopycnal (constant density levels) in the open, stratified ocean, terrain-following coordinates in shallow coastal areas and z-levels in the mixed layer or unstratified deep ocean. See figure below (http://www.hycom.org/hycom/overview).

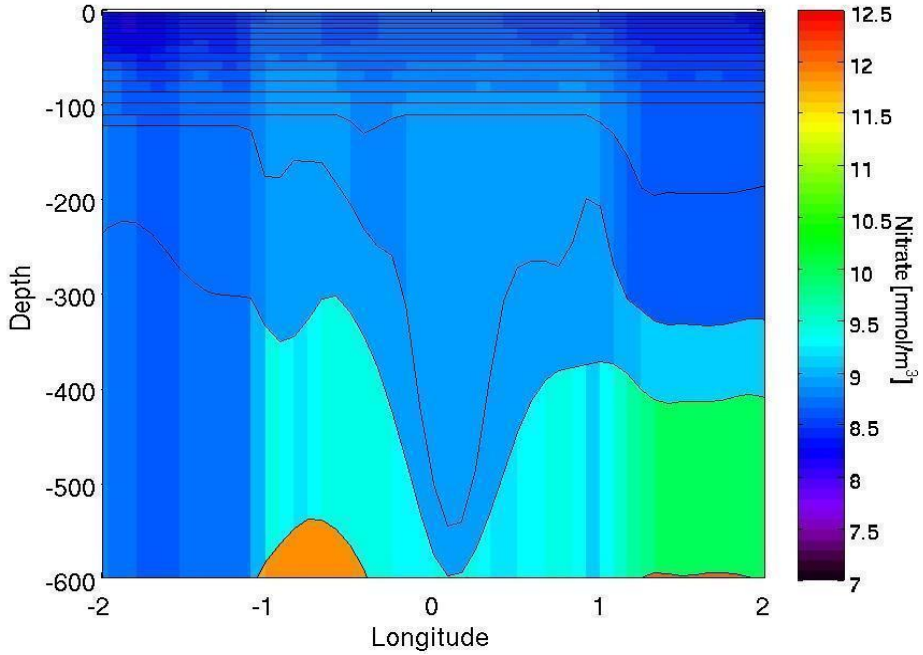


Figure 6 Vertical section of nitrates and vertical levels. Shading indicates concentrations, thin lines indicate layer interfaces. The coordinates revert from isopycnal to z-levels in the mixed layer.

The BGC products are interpolated to 40 levels (different from the PHY model): 0, 2, 3, 4, 5, 6, 8, 10, 11, 13, 16, 18, 22, 25, 29, 34, 40, 47, 56, 66, 78, 92, 110, 131, 156, 186, 222, 266, 318, 380, 454, 541, 644, 763, 902, 1062, 2000, 3000, 3500, 4000 m.

The vertical interpolation method from hybrid layers to fixed levels is a cubic spline honouring the following constraints:

- Conservation of the mean value in each layer
- Continuity and derivability of the value at each layer interface
- Vertical derivatives at the uppermost and lowermost model layers.

The model isopycnic layers can become thin at the bottom of the water column (called *outcropping* of the layers) and are removed from the interpolation, resulting in a few bottom grid cells being masked intermittently as land cells. If this issue turns out problematic, for example for downscaling, a turnaround is to apply the masked values constantly to the pixels where the flickering occurs. We will provide a definitive fix to this issue in a future upgrade of the system.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

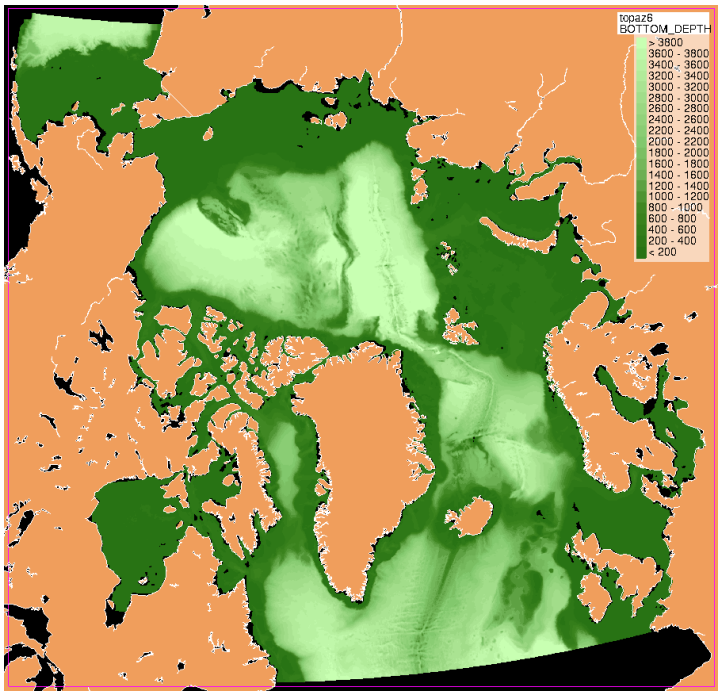
Product Lines	ARCTIC_MULTIYEAR_BGC_002_005
Geographical coverage	North of 63°N and ice-covered North Atlantic Ocean
Variables	• Nitrate • Phosphate • Silicate • Oxygen • Chlorophyll • Phytoplankton biomass • Primary production • Light attenuation • Model depths • zooplankton
Product Type	Multi Year - Reanalysis
Available time series	From 2007 to -2Y
Temporal resolution	24hr average surface 2D fields, monthly and annual average 3D fields
Update Frequency	Yearly
Target delivery time	27th November
Horizontal resolution	25 km
Number of vertical levels	40 levels with a focus on near the surface
Format	NetCDF4

b) Production System Description

The ARC MFC reanalysis uses the NA2a0.80 system based on an advanced sequential data assimilation method (the 1-lag Ensemble Kalman Smoother, EnKS) in its deterministic flavour (DEnKF, Sakov and Oke, 2009) and the Hybrid Coordinate Ocean Model (HYCOM version 2.2.37) coupled online to the ECOSMO-II model (Dœwel and Schrum, 2013)

ARCTIC (180°W-180°E; 63N – 90°N): polar stereographic grid, 216x144 cells

This product is regional with a native grid based on conformal mapping and a very even spatial resolution. It is provided on a standard polar stereographic grid at 25 km resolution. All parameters are interpolated from the native Arakawa C grid model to the same output grid. Vertical interpolation is also applied to the 3D variables in hybrid z-isopycnic layers.



Vertical grid	28 hybrid layers
System Name	NA2a0.80-ECOSMO
Model version	HYCOM v2.2.37 (ocean) CICE 3 (sea ice) ECOSMO II (bgc)
Horizontal output Resolution	25 km
Vertical output Resolution	40 Z levels
Interpolation method	Horizontally: bilinear, vertically: linear
Coupling	Online (subroutines)
Sea Ice Model	CICE v3 with single thickness category

Bathymetry	GEBCO08
Initial conditions	WOA18 nutrients followed by 18 years of spinup.
Lateral boundary conditions	WOA18 climatology (Relaxation in South Atlantic and Bering Strait)
Atmospheric forcings	Source: ECMWF ERA5 Frequency: 6-hourly Resolution: 25 km Other: Completion of solar radiation daily cycle
Atmospheric deposition	None
Tidal constituents	None
Rivers nutrients	GlobalNEWS
Data Assimilation scheme	EnKS (80 dynamic members) + online parameter estimation
Assimilated observations	In situ nutrients (Nitrate, Phosphate, Silicate) BGC Argo chlorophyll-a profiles Level 3 reprocessed Ocean colour (OC CCI product)

The Biogeochemical (BGC) reanalysis system was developed based on the HYCOM-CICE-ECOSMO II ocean-ice-biogeochemical coupled model. The ocean model is based on HYCOM 2.2.37 and its configuration is the 25 km North Atlantic domain, NA2a0.80. In our implementation of HYCOM, the vertical coordinate is isopycnal in the stratified open ocean and z-coordinates in the unstratified surface mixed layer. The data assimilation system for the joint parameter-state estimate is based on the lag-1 ensemble Kalman Smoother system (EnKS).
ECOSMO II (Dæwel and Schrum 2013) is coupled online to the physical HYCOM model; it uses the HYCOM facility for advection of tracers and has the same time-step as the physical model.

Assimilated data

The 3D reanalysis system uses weakly coupled physical and biogeochemical data assimilation. Assimilated data are in-situ biogeochemical data: Nitrate, Phosphate, Silicate and satellite chlorophyll-a (Chl-a) data. Sources of the in-situ data are: 1. Institute of Marine Research nutrient data, 2. GLODAPv2_2019 (Olsen et al., 2019), 3. ICES oceanographic database 4. CLIVAR and 5. CMEMS INS TAC. All in-situ data are quality controlled and duplication of data from multiple data source are merged through TOPAZ preprocessor. The satellite Chl-a data are ESA Ocean Color CCI (OC-CCI) ver 6.0 product (Sathyendranath et al., 2020). Specific OC-CCI product we use here are the global, daily data with 4 km horizontal resolution. The satellite Chl-a data are assumed to be sampled from the ocean surface. In-situ biogeochemical data and satellite Chl-a data are temporally aggregated at 7 daily analysis dates to reduce missing data due to clould. All data are pre-processed to be compatible with model resolution before they are assimilated.
Quality control procedures and preprocessing steps include a range check and horizontal superobing.

The Model

ARCTIC_MULTIYEAR_BGC_002_005 uses version 2.2.37 of HYCOM.
Coverage model: The model domain covers the Nordic Seas and Arctic basin until the Bering Strait, with the horizontal model grid created by a conformal mapping with the poles shifted to the opposite side of the globe to achieve a quasi-homogeneous grid size [Bentsen et al., 1999].

Atmospheric forcing: The reanalysis experiment presented in this document is forced at the ocean surface with fluxes derived from 6-hourly reanalysed atmospheric fluxes from ERA-5 [Dee et al., 2016] that has a resolution of 0.25 deg◦. The atmospheric fields from ERA-5 include: precipitation, dew point temperature, total cloud cover, air temperature at 2 m, sea level pressure, wind speed at 10 m and long wave radiation at the sea surface. The incoming short wave radiation is computed every 3h from synoptic cloud fields, and the wind stress is derived from 10 m winds, estimated as in Large and Pond [1981]. The surface fluxes are forced with a bulk formula parametrisation [Kara, 2000].

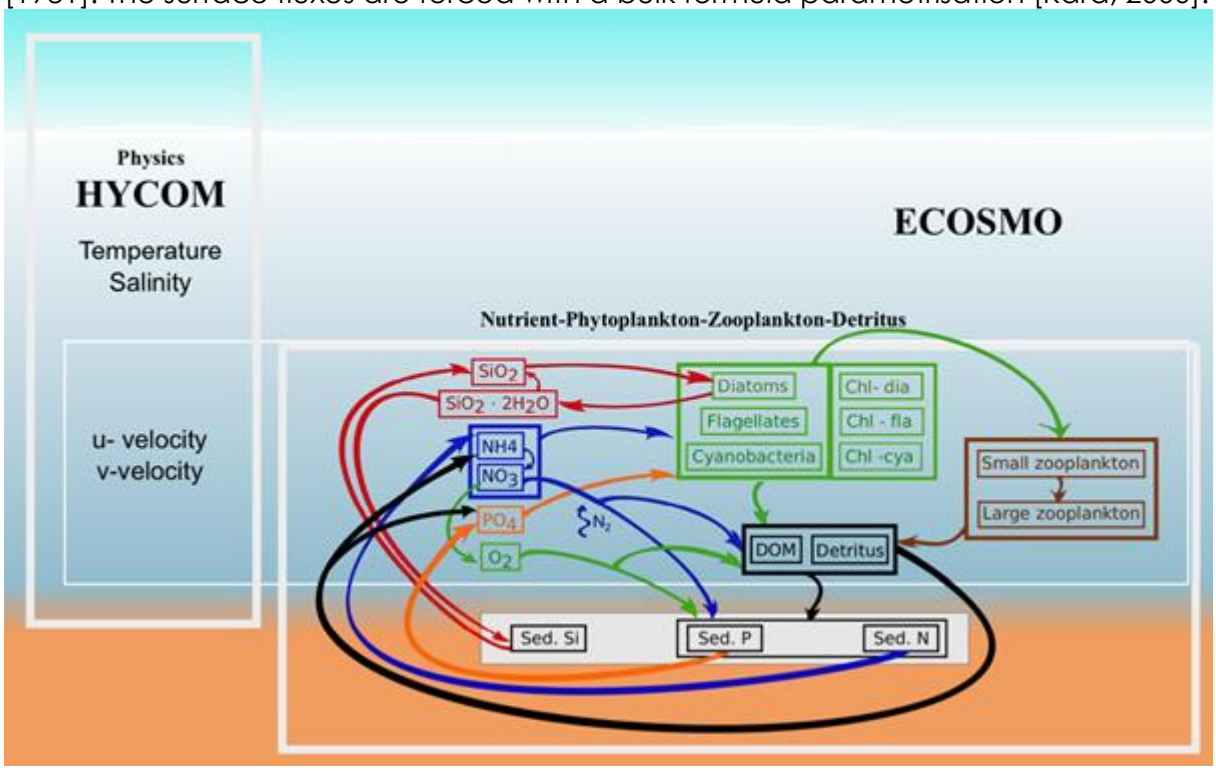


Figure 1: Overview over the components of ECOSMO and how they interact with each other after Yumruktepe et al. (2022).

<u>Data assimilation</u>
In the reanalysis system, surface Chlorophyll data, in-situ nutrients (nitrate, silicate and phosphate) are assimilated into HYCOM-CICE-ECOSMO-II, ice-ocean-biogeochemistry coupled ocean circulation model, at every reanalysis cycle with an 7 day assimilation window. The assimilation system is the lag-1 Ensemble Kalman Smoother (EnKS) with joint state and parameter estimation.
History of the reanalysis: The reanalysis production is conducted independently year by year following a deterministic spin-run from 1987 January 1st and a one year ensemble spin-up run starting from January 1st the year before.
The EnKS estimates also scalar biological parameters of the ECOSMO model.

c) Dataset details

Product Lines	ARCTIC_ANALYSISFORECAST_BGC_002_005	
Geographical coverage	Arctic	
Datasets: cmems_mod_arc_bgc_my_ecosmo_P1D-m cmems_mod_arc_bgc_my_ecosmo_P1M cmems_mod_arc_bgc_my_ecosmo_P1Y		
Product Type	Reanalysis	
Update frequency	Yearly	
Ensemble	Average of 80 members	
Available time series	2007 – Y-2 with PHY+full BGC DA	Y
Target delivery time	November	
Temporal resolution (respective datasets)	Daily mean Monthly mean Yearly mean	

Datasets: cmems_mod_arc_bgc_anfc_ecosmo_P1D-m **cmems_mod_arc_bgc_anfc_ecosmo_P1M-m** Contain the 3D variables			
Variable	**Unit**	**Name in file**	**Standard name**
Nitrate	mmol m-3	no3	mole_concentration_of_nitrate_in_sea_water
Phosphate	mmol m-3	po4	mole_concentration_of_phosphate_in_sea_water
Oxygen	mmol m-3	o2	mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
Primary production	mg m-3 day-1	nppv	net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
Chlorophyll	mg m-3	chl	mass_concentration_of_chlorophyll_a_in_sea_water
Silicate	mmol m-3	si	mole_concentration_of_silicate_in_sea_water
Phytoplankton biomass	mmol m-3	phyc	mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
Light attenuation	m-1	kd	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Model depth	m	model_depth	sea_floor_depth_below_sea_level

Zooplankton biomass	mmol m-3	zooc	mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water

d) Additional information on some parameters

None.

e) Processing information

The reanalysis production is conducted independently year by year following a deterministic spin-run from 2000 January 1st and a one year ensemble spin-up run starting from January 1st the year before.

f) Catalogue updating

The reanalysis is updated every year with biogeochemical (BGC) data assimilation until the Y - 2 in order to benefit from reprocessed in situ BGC data in the Arctic.

g) Time and ensemble averaging

All provided variables are averages of the 80-members ensemble.
Daily averages are centered at 12:00 UTC.
Monthly averages are centered at the middle of the months.

h) Product use and impact

The Arctic BGC reanalysis uses a more advanced and more expensive data assimilation method compared to the corresponding NRT product and is thus run at much coarser resolution. The best compromise between resolution and advanced data assimilation method is not yet known in terms of accuracy, realism and forecast skills.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The ARC MFC products have been set up through different phases of Copernicus Marine and the file and dataset naming conventions have changed in the meantime. The table 2 below shows correspondence between product name, dataset name, and file name.

Table 2: File name convention.

Product Name	Dataset Name	File Name
ARCTIC_MULTIYEAR_BGC_002_005	cmems_mod_arc_bgc_my_ecosmo_P1D-m	20070330_dm-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc
	cmems_mod_arc_bgc_my_ecosmo_P1M-m	20070315_mm-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc
	cmems_mod_arc_bgc_my_ecosmo_P1Y	2007_ym-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [MB]
cmems_mod_arc_bgc_my_ecosmo_P1D-m	20070330_dm-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc	16.5
cmems_mod_arc_bgc_my_ecosmo_P1M-m	20070315_mm-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc	444
cmems_mod_arc_bgc_my_ecosmo_P1Y	2007_ym-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0.nc	444

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

5) REFERENCES

Yumruktepe, V. Ç., Samuelsen, A., and Daewel, U.: ECOSMO II(CHL): a marine biogeochemical model for the North Atlantic and the Arctic, Geosci. Model Dev., 15, 3901–3921, https://doi.org/10.5194/gmd-15-3901-2022, 2022.

Quality Information Document:

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-ARC-QUID-002-005.pdf

ANNEX

Sample file: *netcdf \20070101_dm-25km-NERSC-MODEL-ECOSMO-ARC-RAN-fv2.0{dimensions:*

```
        time = UNLIMITED ; // (1 currently)
        x = 297 ;
        y = 285 ;
        depth = 1 ;
variables:
        double time(time) ;
                time:long_name = "forecast time" ;
                time:units = "hour since 1950-1-1T00:00:00Z" ;
        int stereographic ;
```

```
                stereographic:grid_mapping_name = "polar_stereographic" ;
                stereographic:latitude_of_projection_origin = 90. ;
                stereographic:longitude_of_projection_origin = -45. ;
                stereographic:scale_factor_at_projection_origin = 1. ;
                stereographic:straight_vertical_longitude_from_pole = -45. ;
                stereographic:earth_radius = 6378273. ;
                stereographic:proj4 = "+proj=stere +lon_0=-45 +lat_0=90 +k=1 +R=6378273
+no_defs" ;
                stereographic:false_easting = 0. ;
                stereographic:false_northing = 0. ;
        float x(x) ;
                x:axis = "X" ;
                x:standard_name = "projection_x_coordinate" ;
                x:units = "100  km" ;
        float longitude(y, x) ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
        float y(y) ;
                y:standard_name = "projection_y_coordinate" ;
                y:axis = "Y" ;
                y:units = "100  km" ;
        float latitude(y, x) ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
        float depth(depth) ;
                depth:long_name = "depth" ;
                depth:units = "m" ;
                depth:standard_name = "depth" ;
                depth:positive = "down" ;
                depth:axis = "Z" ;
        short model_depth(y, x) ;
                model_depth:_FillValue = -32767s ;
                model_depth:missing_value = -32767s ;
                model_depth:add_offset = 5000.5 ;
                model_depth:scale_factor = 0.152626438360346 ;
                model_depth:units = "meter" ;
                model_depth:standard_name = "sea_floor_depth_below_sea_level" ;
                model_depth:grid_mapping = "stereographic" ;
                model_depth:coordinates = "longitude latitude" ;
                model_depth:cell_methods = "area: mean" ;
        short chl(time, depth, y, x) ;
                chl:_FillValue = -32767s ;
                chl:missing_value = -32767s ;
                chl:add_offset = 50. ;
                chl:scale_factor = 0.00152611177242621 ;
                chl:units = "mg m-3" ;
                chl:standard_name =
"mass_concentration_of_chlorophyll_a_in_sea_water" ;
                chl:grid_mapping = "stereographic" ;
```

```
                chl:coordinates = "longitude latitude" ;
                chl:cell_methods = "area: mean" ;
        short phyc(time, depth, y, x) ;
                phyc:_FillValue = -32767s ;
                phyc:missing_value = -32767s ;
                phyc:add_offset = 50. ;
                phyc:scale_factor = 0.00152611177242621 ;
                phyc:units = "mmol m-3" ;
                phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
                phyc:grid_mapping = "stereographic" ;
                phyc:coordinates = "longitude latitude" ;
                phyc:cell_methods = "area: mean" ;
        short zooc(time, depth, y, x) ;
                zooc:_FillValue = -32767s ;
                zooc:missing_value = -32767s ;
                zooc:add_offset = 50. ;
                zooc:scale_factor = 0.00152611177242621 ;
                zooc:units = "mmol m-3" ;
                zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water" ;
                zooc:grid_mapping = "stereographic" ;
                zooc:coordinates = "longitude latitude" ;
                zooc:cell_methods = "area: mean" ;
        float o2(time, depth, y, x) ;
                o2:_FillValue = -1.e+14f ;
                o2:missing_value = -1.e+14f ;
                o2:units = "mmol m-3" ;
                o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
                o2:grid_mapping = "stereographic" ;
                o2:coordinates = "longitude latitude" ;
                o2:cell_methods = "area: mean" ;
        short no3(time, depth, y, x) ;
                no3:_FillValue = -32767s ;
                no3:missing_value = -32767s ;
                no3:add_offset = 17.5 ;
                no3:scale_factor = 0.000534139120349174 ;
                no3:units = "mmol m-3" ;
                no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
                no3:grid_mapping = "stereographic" ;
                no3:coordinates = "longitude latitude" ;
                no3:cell_methods = "area: mean" ;
        short po4(time, depth, y, x) ;
                po4:_FillValue = -32767s ;
                po4:missing_value = -32767s ;
                po4:add_offset = 1. ;
                po4:scale_factor = 3.05222354485242e-05 ;
                po4:units = "mmol m-3" ;
```

```
                po4:standard_name =
"mole_concentration_of_phosphate_in_sea_water" ;
                po4:grid_mapping = "stereographic" ;
                po4:coordinates = "longitude latitude" ;
                po4:cell_methods = "area: mean" ;
        short si(time, depth, y, x) ;
                si:_FillValue = -32767s ;
                si:missing_value = -32767s ;
                si:add_offset = 17.5 ;
                si:scale_factor = 0.000534139120349174 ;
                si:units = "mmol m-3" ;
                si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
                si:grid_mapping = "stereographic" ;
                si:coordinates = "longitude latitude" ;
                si:cell_methods = "area: mean" ;
        short kd(time, depth, y, x) ;
                kd:_FillValue = -32767s ;
                kd:missing_value = -32767s ;
                kd:add_offset = 0.25 ;
                kd:scale_factor = 7.63055886213106e-06 ;
                kd:units = "m-1" ;
                kd:standard_name =
"volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water" ;
                kd:grid_mapping = "stereographic" ;
                kd:coordinates = "longitude latitude" ;
                kd:cell_methods = "area: mean" ;
        short nppv(time, depth, y, x) ;
                nppv:_FillValue = -32767s ;
                nppv:missing_value = -32767s ;
                nppv:add_offset = 100. ;
                nppv:scale_factor = 0.00305222354485243 ;
                nppv:units = "mg m-3 day-1" ;
                nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_
water" ;
                nppv:grid_mapping = "stereographic" ;
                nppv:coordinates = "longitude latitude" ;
                nppv:cell_methods = "area: mean" ;

// global attributes:
                :title = "Arctic Ocean Biogeochemistry Reanalysis, 25km surface daily
mean" ;
                :institution = "NERSC, Jahnebakken 3, 5006 Bergen, Norway" ;
                :source = "NERSC-HYCOM model fields" ;
                :references = "http://topaz.nersc.no" ;
                :field_type = "Files based on file type nersc_daily" ;
                :Conventions = "CF-1.4" ;
                :field_date = "2007-01-01" ;
```

:NCO = "netCDF Operators version 4.7.9 (Homepage = http://nco.sf.net, Code = http://github.com/nco/nco)" ;
:history = "Created by program hyc2proj, version V0.3" ;